Status	Finished
Started	Friday, 30 October 2026, 2:50 PM
Completed	Friday, 30 October 2026, 2:53 PM
Duration	3 mins 43 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 10 ✓
- b. 500
- c. 1000
- d. 100

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list is sorted ✓
- b. When the list contains duplicate elements
- c. When the list is unsorted
- d. When the list is very small

Question **3**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Inserting elements into a list
- b. Deleting elements from a list
- c. Sorting elements in a list
- d. Determining whether an element is present in a list ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It is slower than linear search for small lists

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search divides the list into halves.
b. Linear search requires the list to be sorted.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search is more efficient than binary search on large lists.

Question **6**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 94
b. 90 and 94
c. 90 and 99 ✓
d. 89 and 99

Question **7**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the list contains duplicate elements
c. When the search algorithm finishes
d. When the element is found in the list ✓

Question **8**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all

b. When the item is somewhere in the middle of the array ✓

c. When the item is the last element in the array

d. When the item is not the array at all

Question **9**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts

b. Compare the target element with the first element in the list

c. Sort the list

d. Compare the target element with the middle element in the list ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)

b. O(log n)

c. O(1)

d. O👎 ✓

Question **11**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. An error is raised

b. The last element is returned

c. The first element is returned

d. The search is considered unsuccessful ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and unsorted
b. When the list is large and sorted
c. When the list is small and unsorted ✓
d. When the list is small and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)
b. Binary ✓
c. Linear
d. Hash

Question **14**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. At the beginning of the list
b. In the middle of the list
c. In the left sublist
d. In the right sublist ✓

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search has a lower time complexity on large, sorted lists ✓